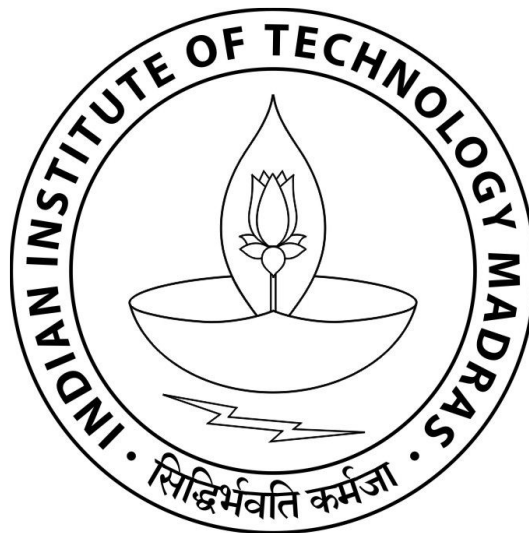


Barbell Grip: Moisture, Chalk, and Slip Risk

Choose Focus Analysis Exercise

BT5051: Transport Phenomena in Biological Systems



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BE23B042

3rd November, 2025

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1. Introduction

Grip reliability is a practical safety and performance concern in strength training. As a set progresses, perspiration accumulates at the hand–bar interface, forming a thin moisture film that can substantially reduce friction and increase the likelihood of micro-slip or bar roll. Athletes commonly apply magnesium carbonate (chalk) to absorb moisture and improve feel, yet chalking practices are often ad hoc and sensitive to room humidity, airflow, bar knurling, and individual sweat rates. This project develops a concise, first-principles model to quantify how sweat, evaporation, and chalk jointly determine the time window over which a secure grip can be maintained.

The core idea is straightforward: the friction coefficient μ at the palm–bar interface is strongly dependent on the interfacial moisture state. Moisture increases through sweat secretion and decreases through two sinks—evaporation to room air and sorption into a finite-capacity chalk layer. When the moisture film is thin (dry regime), asperity interlock and adhesion support higher friction; as the film thickens (boundary-lubricated regime), μ drops and slip risk rises. Chalk improves the margin until it becomes saturated, after which its effectiveness declines.



Figure 1. Preparing hands with chalk (magnesium carbonate), a common strategy to reduce moisture and improve friction at the palm–bar interface.

Image source : <https://share.google/images/jtEkwZwcaedFq4jmo>



Figure 2. Illustration of the CG-MTP (clean-grip mid-thigh pull), a heavy barbell movement where grip reliability is critical.

Image source : <https://share.google/images/85PFB8SCzs0NevytC>

This report treats the hand–bar contact as a small, representative area and uses standard ideas from Continuum Analysis/Transport Phenomena—mass conservation, fluxes, and transfer-coefficient methods—to turn the above narrative into a compact, quantitative description of grip behavior during a set. With transparent, gym-realistic parameters (temperature, RH, local air speed, sweat rate, chalk amount, and normal force), we track how the moisture film evolves and how that state maps to friction and a simple safety margin. The focus is intentionally practical: to make clear, reproducible plots that show how long a secure grip lasts under typical gym room conditions, how ventilation or humidity shift that window, and why timely re-chalking helps. By grounding a familiar training issue in first principles, the analysis connects classroom concepts to a decision a lifter makes every day at the platform.

2. Motivation

Strength training has been a steady interest of mine, even though the last few semesters have been difficult. Between academic load and my father's blood cancer diagnosis in second semester, I stopped going to the gym regularly. The interest never went away, but time and motivation did. When I did manage NSO sessions at the IIT Madras weightlifting area, I kept running into the same practical problem: after just two or three reps, my palms would sweat, the knurled bar would feel slick, and the weight became harder to control. Applying chalk helped immediately—but I was guessing when to use it and how much.

This CFA gives me a constructive way to turn that lived experience into a clear, quantitative understanding. Instead of relying on trial and error, I want to explain why the grip “falls off” mid-set and how simple actions—re-chalking at the right time, improving airflow with a small fan, or choosing a different knurl—can prevent it. The topic also keeps me connected to the gym even while academics and family responsibilities limit training time; it channels my interest into analysis that can help me and others lift more safely.

The problem aligns naturally with this course. The moisture at the hand–bar interface can be treated with mass conservation (sweat in; evaporation and chalk absorption out) and transfer-coefficient ideas for air-side removal. A simple constitutive relation links that moisture state to friction, allowing us to predict when available friction drops below what the lift requires. These are exactly the tools emphasized in Continuum Analysis/Transport Phenomena, applied to a familiar, real-world setting.

Looking ahead, I want to become an AI/ML engineer, and my Nirmaan internships have trained me to turn intuition into code, structure messy problems, and validate models against reality. I will bring that mindset to this CFA—use transparent parameters, write a reproducible Python script, run sensitivity sweeps, and convert the outputs into simple, gym-facing recommendations. In that sense, this project is both a personal problem I care about and a small applied modeling exercise that builds the skills I aim to use in my career.

3. Objective

- Build a small, physics-based model of barbell grip that tracks moisture at the hand–bar interface from sweat, evaporation (airflow), and chalk.
- Link moisture to friction with a simple $\mu(h)$ relation so we can estimate grip strength over time.
- Predict the no-slip window (time/reps before failure) and the best re-chalk interval under typical gym conditions (humidity, airflow, knurl, load).

4. Concepts used

4.1 Conservation of mass

A mass balance on a small control area A at the palm–bar interface. A thin liquid film of thickness $h(t)$ covers this area; the areal mass of water is $\rho_w h (\text{kg} \cdot \text{m}^{-2})$. One source adds water (sweat from eccrine glands), and two sinks remove it (evaporation to the room air and absorption into chalk).

General form:

In + Generation – Out – Consumption = Accumulation. (4.1.1)

$$\mathbf{I + G - O - C = A}$$

Our film (areal form): one source (sweat) and two sinks (evaporation, chalk).

$$\frac{d(\rho_w h)}{dt} = J_{\text{sweat}} - J_{\text{evap}} - J_{\text{chalk}} \Rightarrow \boxed{\frac{dh}{dt} = \frac{1}{\rho_w} (J_{\text{sweat}} - J_{\text{evap}} - J_{\text{chalk}})} \quad (4.1.2)$$

- $h(t)$: film thickness (m), ρ_w : water density ($\text{kg} \cdot \text{m}^{-3}$), J : flux ($\text{kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$).
This ODE is the backbone of the model.

This ODE is the core of our model; later sections just define each flux. We assume a uniform thin film over area A (lumped model), which is reasonable for short sets focused on when slip begins, not where. We start nearly dry: $h(0) = h_0 \ll h_c$.

4.2 Mass flux & Fick's idea

Mass (or molar) flux is Amount crossing unit area per time, normal to the surface. For diffusion through a thin layer:

$$\boxed{J = -D \frac{\partial c}{\partial y}} \quad (4.2)$$

- D : diffusivity (m^2/s)
- c : concentration

- y : normal direction

where J is flux, D diffusivity (m^2/s), c concentration, and y the normal coordinate. With a finite gas-side boundary layer of thickness δ :

$$J \approx D \frac{c_{v,s} - c_{v,\infty}}{\delta} = k_m (c_{v,s} - c_{v,\infty}),$$

$$k_m \equiv D/\delta.$$

I assume the gas side controls and the liquid film is well mixed—a standard evaporation simplification—so we can estimate k_m from airflow/geometry (via Re , Sc , Sh).

4.3 Transfer-coefficient model for evaporation

Following the course's k -type approach, we express evaporation to room air as

$$J_{\text{evap}} = k_m (c_{v,s} - c_{v,\infty}), \quad (4.3)$$

with k_m the air-side mass-transfer coefficient (m/s). Concentrations are obtained from the ideal-gas relation $c_v = \frac{M_w}{RT} p_v$. For a surface at temperature T ,

$$c_{v,s} = \frac{M_w}{RT} p_{ws}(T), \quad c_{v,\infty} = \frac{M_w}{RT} \phi p_{ws}(T), \quad (4.4)$$

where $p_{ws}(T)$ is saturation vapor pressure and ϕ is relative humidity. Thus high ϕ shrinks the driving force, slowing drying. To estimate k_m we use a standard Sherwood correlation for external flow over a small patch (analogy to Nusselt for heat):

$$Sh = \frac{k_m L}{D_v} \approx 0.664 Re^{1/2} Sc^{1/3} \Rightarrow k_m \approx 0.664 \frac{D_v}{L} Re^{1/2} Sc^{1/3}, \quad (4.5)$$

with $Re = uL/\nu$ and $Sc = \nu/D_v$. Here u is local air speed near the hand, L a characteristic hand width, ν air kinematic viscosity, and D_v the diffusivity of water vapor in air. This correlation makes the model actionable: a small fan increases

u , which increases Re and therefore k_m , enlarging the no-slip window. In code, k_m is computed once per scenario (constant T, φ, u during a set).

4.4 Chalk as a capacity-limited porous sink

Chalk (magnesium carbonate) is treated as a thin porous layer that absorbs liquid water until full.

- M_{cap} : max water per area the chalk can hold ($\text{kg}\cdot\text{m}^{-2}$).
- $m_c(t)$: water already stored per area; $\theta = m_c/M_{\text{cap}} \in [0,1]$ (saturation).

Uptake law (lumped):

$$\boxed{J_{\text{chalk}} = k_s h (1 - \theta)}, \quad \boxed{\frac{d\theta}{dt} = \frac{J_{\text{chalk}}}{M_{\text{cap}}}} \quad (4.6)$$

- $k_s (\text{s}^{-1})$ bundles porosity/permeability/contact effects.
As $\theta \rightarrow 1$, uptake $\rightarrow 0$: saturated chalk stops helping.

4.5 Friction–moisture relation

We model friction as a smooth drop from “dry” to “wet” with film thickness h :

$$\mu(h) = \mu_{\text{wet}} + \frac{\mu_{\text{dry}} - \mu_{\text{wet}}}{1 + (h/h_c)^n} \quad (4.7)$$

- $\mu_{\text{dry}}, \mu_{\text{wet}}$: friction limits (dry > wet).
- h_c : characteristic thickness where μ is halfway between dry and wet.
- n : steepness of the transition.
- N : normal force; available grip $F_{\text{avail}} = \mu(h)N$ (Coulomb).

thin films ($h \ll h_c$) $\rightarrow \mu \approx \mu_{\text{dry}}$;

thicker films ($h \gg h_c$) $\rightarrow \mu \approx \mu_{\text{wet}}$.

Squeezing harder raises μN , but only drying (chalk/airflow) moves μ back toward the dry, high-friction regime.

4.6 Failure (no-slip criterion and safety margin)

Available friction must exceed the lift's tangential demand:

$$F_{\text{avail}} = \mu(h(t)) N \quad (4.8)$$

Define a safety margin:

$$\boxed{S(t) = \frac{\mu(h(t)) N}{F_{\text{req}}}} \text{ fail when } S(t) \leq S_{\text{min}}. \quad (4.9)$$

We report the no-slip window as the time (or reps) until the threshold is crossed.

4.7 Non - dimensionalization (regime map for design)

Choose moisture scale h_c and an evaporation time scale

$$t_c = \frac{\rho_w h_c}{k_m \Delta c^*}, \Delta c^* = \frac{M_w}{RT} (p_{ws}(T_s) - \varphi p_{ws}(T_\infty)).$$

Let $h^* = h/h_c$, $t^* = t/t_c$. Then (4.1), (4.5)–(4.6) give

$$\boxed{\frac{dh^*}{dt^*} = \frac{J_{\text{sweat}}}{k_m \Delta c^*} - 1 - \frac{k_s h_c}{k_m \Delta c^*} h^* (1 - \theta)} \quad (4.10)$$

$\check{\Pi}_s \qquad \check{\kappa}$

$$\boxed{\frac{d\theta}{dt^*} = \frac{k_s \rho_w h_c}{M_{\text{cap}}} h^* (1 - \theta)} \quad (4.11)$$

$\check{\beta}$

The groups mean (simple):

- Π_s — sweat input / evaporation capacity (Damköhler-like).
 - $\Pi_s \ll 1$: evaporation alone keeps the grip dry.

- $\Pi_s \gtrsim 1$: need chalk or stronger airflow.
- κ — chalk strength / evaporation strength.
 - Large κ : fresh chalk removes moisture quickly.
- β — how fast chalk saturates.
 - Large β : chalk fills fast $\rightarrow$ re-chalk more often.

These groups drive our regime maps and the re-chalk timing chart.

5. Overview of Relevant Information

5.1 Grip chalk in clean-grip mid-thigh pull (CG-MTP)

A study on high-school athletes compared CG-MTP performance at 70% 1-RM with and without magnesium carbonate. In a repeated-measures design, lifters completed more repetitions with chalk ($\approx 11.1 \pm 5.1$) than without ($\approx 9.1 \pm 3.1$), a statistically significant $\sim 22.5\%$ gain. The authors recommend chalk for lifts where grip can limit output. for our CFA, this supports treating chalk as a practical intervention that extends the “no-slip window,” which aligns with our plan to predict a re-chalk interval.

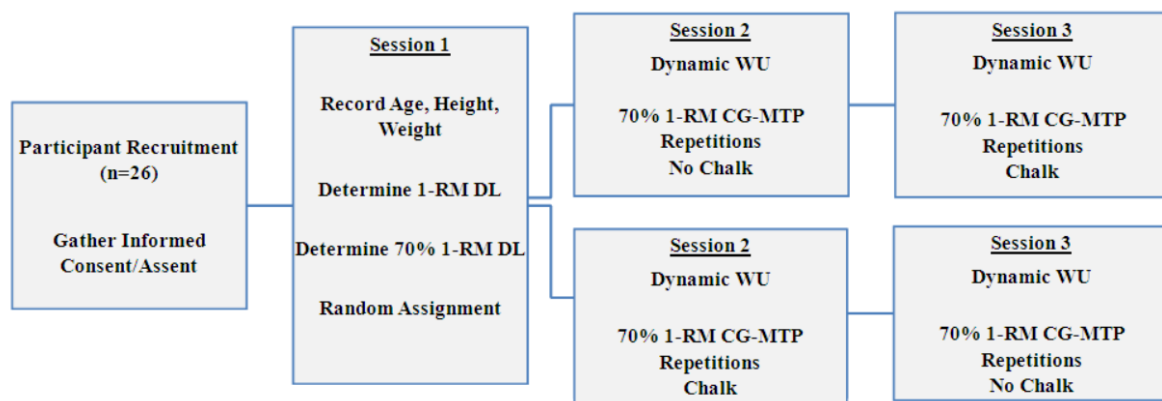


Figure 5.1. Study timeline of events. 1-RM one repetition maximum. CG-MTP closed grip midhigh pull

Image source :- <http://article.sapub.org/10.5923.j.sports.20201002.04.html>

5.2 Chalk as a friction modifier at the finger–surface interface

Fingertip tribology shows that chalk’s effect is context-dependent: it often helps in dry conditions by filling asperity gaps ($\uparrow$ real contact), but in wet conditions excess chalk can act as a sheared third body and lower friction; chalk also reduces variability, making grip more predictable. Skin friction arises from adhesion (real contact area) and deformation/ploughing; moisture first softens skin (can raise friction) and, when excessive, forms a lubricating film (lowers friction). We map these mechanisms into the model via film thickness $h(t)$, chalk saturation $\theta(t)$, and a moisture–friction law $\mu(h)$. The source is sweat flux J_{sweat} (grows h), while sinks are evaporation $J_{\text{evap}} = k_m(c_{v,s} - c_{v,\infty})$ (stronger with airflow; weaker at high RH) and chalk uptake $J_{\text{chalk}} = k_s h(1 - \theta)$ (effective until capacity M_{cap}). Design levers follow: improve ventilation ($\uparrow k_m$), manage

chalk freshness/amount ($\uparrow$ effective k_s , capacity), choose suitable knurl, and adjust normal force N . Accordingly, our figures report time-to-slip curves ($h(t)$, $\mu(h(t))$, margin $S(t)$), a re-chalk interval vs humidity/airflow/load, and sensitivities to knurl/roughness proxies and N —then we check that the trends align with CG-MTP results (more reps with chalk) and tribology findings (context-dependent benefit, improved consistency).

REFERENCE LINK : <https://eprints.whiterose.ac.uk/id/eprint/214893/10/clarke-et-al-2024-the-effectiveness-of-chalk-as-a-friction-modifier-for-finger-pad-contact-with-rocks-of-varying.pdf>

6. Approach

What we do. We build a small, reproducible simulation that tracks the water film at the palm–bar interface during a set, converts that moisture state into a friction coefficient, and reports the time (or reps) until slip risk crosses a chosen safety margin. All equations come from Section 4, and we keep parameters explicit so the plots can be regenerated for general gym conditions.

6.1 System definition and inputs

We model one hand's effective contact patch A on a knurled steel bar. The state variables are: film thickness $h(t)$ (m) and chalk saturation $\theta(t) \in [0,1]$. External inputs are:

- **Environment:** air temperature T (K), relative humidity φ (0–1), local air speed over the hand u (m·s⁻¹).
- **Geometry/transport:** characteristic length L of the patch (~hand width), vapor diffusivity D_v , air kinematic viscosity ν .
- **Chalk:** effective uptake rate k_s (s⁻¹), areal capacity M_{cap} (kg·m⁻²), initial saturation θ_0 .
- **Sweat:** flux J_{sweat} (kg·m⁻²·s⁻¹); piecewise-constant (baseline → higher during the set).
- **Mechanics:** normal force N (N) and tangential demand F_{req} (N) representative of the lift phase we care about (steady hold or the “hardest” part of the pull).

We start each scenario with $h(0) = h_0$ (nearly dry after chalking/wiping) and $\theta(0) = \theta_0$ (fresh or partially used chalk).

6.2 Governing equations (implemented exactly as written)

The moisture balance on the control area is

$$\frac{dh}{dt} = \frac{1}{\rho_w} (J_{\text{sweat}} - J_{\text{evap}} - J_{\text{chalk}}), \rho_w \simeq 1000 \text{ kg}\cdot\text{m}^{-3}.$$

└─ Constant film density ρ_w (liquid water, ~1000 kg·m⁻³)

Evaporation uses the transfer-coefficient form

$$J_{\text{evap}} = k_m (c_{v,s} - c_{v,\infty})$$

$$k_m \approx 0.664 \frac{D_v}{L} \text{Re}^{\frac{1}{2}} \text{Sc}^{\frac{1}{3}}$$

$$\text{Re} = \frac{uL}{\nu}, \text{Sc} = \frac{\nu}{D_v}.$$

With ideal-gas relations:

$$c_{v,s} = \frac{M_w}{RT} p_{ws}(T)$$

$$c_{v,\infty} = \frac{M_w}{RT} \phi p_{ws}(T).$$

Chalk uptake is a capacity-limited sink:

$$J_{\text{chalk}} = k_s h (1 - \theta), \frac{d\theta}{dt} = \frac{J_{\text{chalk}}}{M_{\text{cap}}}.$$

Grip friction follows a smooth dry→wet relation:

$$\mu(h) = \mu_{\text{wet}} + \frac{\mu_{\text{dry}} - \mu_{\text{wet}}}{1 + (h/h_c)^n}.$$

Safety margin and failure time:

$$S(t) = \frac{\mu(h(t)) N}{F_{\text{req}}}, \text{declare "failure" when } S(t) \leq S_{\text{min}} \text{ (e.g., 1.0 or 1.2).}$$

6.3 Parameterization for base runs

We use transparent, gym-realistic starting values and then perform sensitivity sweeps:

- **Environment:** $T = 298\text{K}$ (25 °C); $\varphi = 0.6$ (humid room) $\rightarrow$ also test 0.4 and 0.8. Air speed u : still air $0.05 \text{ m}\cdot\text{s}^{-1}$; small fan $0.5 \text{ m}\cdot\text{s}^{-1}$. $L = 0.08\text{m}$; $D_v = 2.5 \times 10^{-5} \text{ m}^2/\text{s}$; $\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$. These give $k_m \approx 1\text{--}9 \times 10^{-3} \text{ m/s}$ (still $\rightarrow$ fan).
- **Sweat:** $J_{\text{sweat}} = 3 \times 10^{-4} \text{ kg}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ during the set ($\pm 3\times$ in sweeps).
- **Chalk:** $k_s = 1.0 \text{ s}^{-1}$ (baseline), $M_{\text{cap}} = 5 \times 10^{-3} \text{ kg}\cdot\text{m}^{-2}$, $\theta_0 = 0$ (fresh) or 0.5 (partially used).
- **Friction law:** $\mu_{\text{dry}} = 0.9$, $\mu_{\text{wet}} = 0.4$, $h_c = 5 \times 10^{-6} \text{ m}$, $n = 3$.
- **Mechanics:** $N = 300\text{--}600 \text{ N}$ per hand (firm squeeze); $F_{\text{req}} = 150\text{--}300 \text{ N}$ per hand (depends on lift and moment arm). We report results as a margin $S(t)$ so users can map to their loads.

All values are editable in code and will be listed in a small table. And the python code I provided in the last of the CFA

6.4 Numerical solution and outputs

We integrate the coupled ODEs for $h(t)$ and $\theta(t)$ with a fixed-step 4th-order Runge–Kutta that I learnt in BT2020 course (time step $\Delta t = 0.05\text{--}0.1 \text{ s}$; horizon 60–180 s). At each step we compute $\mu(h(t))$ and $S(t)$. Primary outputs:

1. **Time traces:** $h(t)$, $\theta(t)$, $\mu(h(t))$, and $S(t)$ (with the failure time marked).
2. **Re-chalk interval:** the earlier of (i) $S(t)$ crossing S_{min} or (ii) chalk saturation reaching $\theta \geq 0.8$.
3. **Design charts:** (a) re-chalk interval vs humidity and fan speed; (b) safe window vs N and F_{req} ; (c) sensitivity of outcomes to k_s and M_{cap} .
4. **Regime map:** contours of the non-dimensional groups Π_s , κ , β showing where evaporation alone suffices, where fresh chalk is effective, and where chalk saturates quickly.

- Baseline film thickness $h(t)$: You can see how quickly moisture builds on the palm–bar patch under baseline gym conditions.

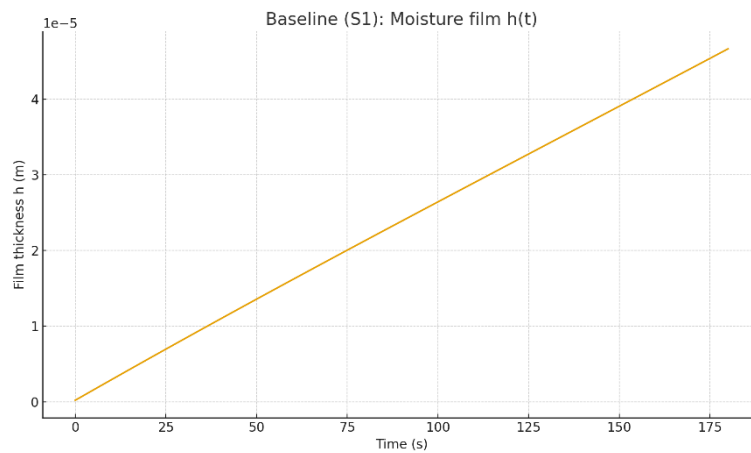


Figure 6.4.1: $h(t)$ starts near-dry and rises as sweat outpaces evaporation, then slows as chalk saturates. This is the upstream driver of all later effects (friction drop, slip risk).

- Baseline chalk saturation $\theta(t)$: Track how fast chalk fills up.

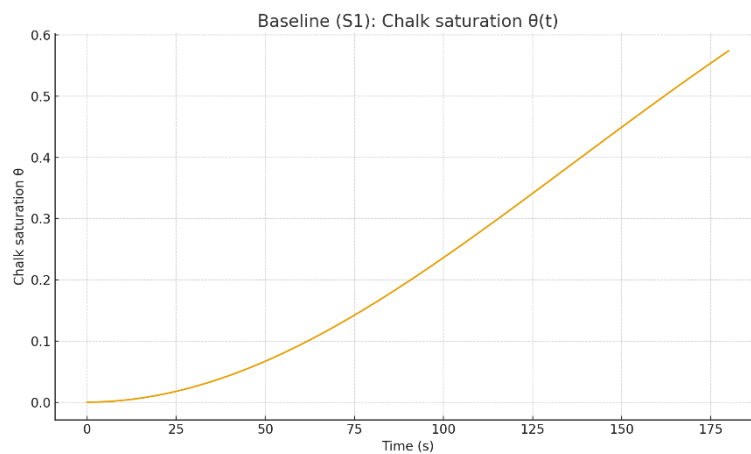


Figure 6.4.2 : Rapid early uptake ($\theta \uparrow$) then a plateau near $\theta \rightarrow 1$. The re-chalk time is chosen before saturation (e.g., $\theta \approx 0.8$), because benefit fades after that.

- Baseline friction $\mu(t)$: Convert moisture into grip quality.

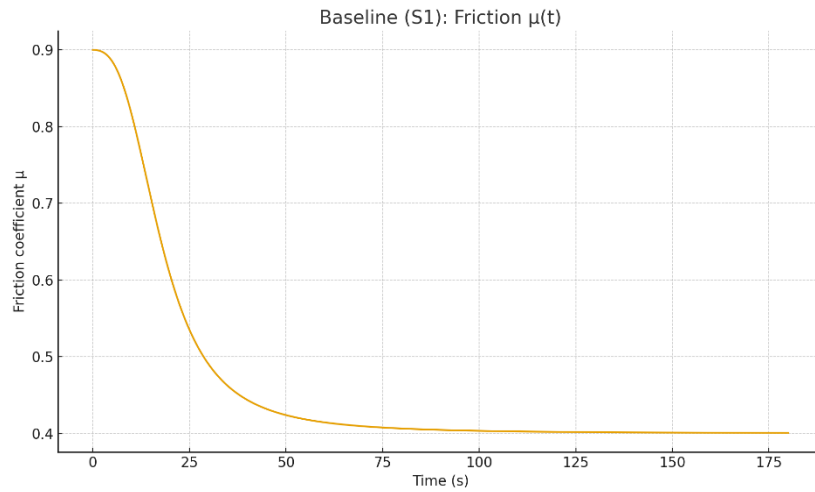


Figure 6.4.3: A smooth fall from μ_{dry} to μ_{wet} as $h(t)$ grows. The time where $\mu(t)$ bends downward corresponds to $h \sim h_c$ (onset of “wet” behavior).

- Baseline safety margin $S(t)$: This is the decision signal.

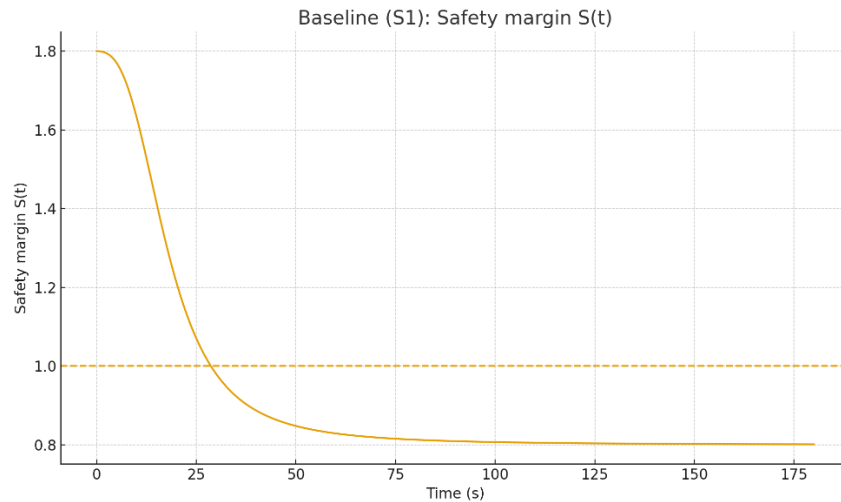


Figure 6.4.4 : $S(t) = \mu(h(t))N/F_{\text{req}}$ declining over time; the **no-slip window** is where $S(t)$ first crosses the threshold (S_{min}). That crossing gives the recommended **re-chalk interval** if earlier than chalk saturation.

- Re-chalk interval vs humidity (still air vs small fan): A design chart for the room environment.

Result : Partially used chalk ($\theta_0 = 0.5$): ≈ 28.5 s.

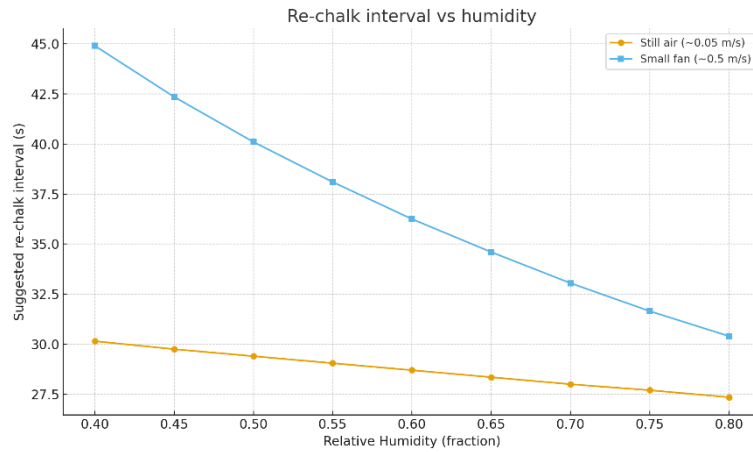


Figure 6.4.5 : Higher RH $\rightarrow$ shorter safe window. A small fan (higher k_m) shifts the curve upward (more seconds before re-chalk). Quantifies the benefit of ventilation on humid days.

- Safe window vs grip force N : How much does squeezing harder help.

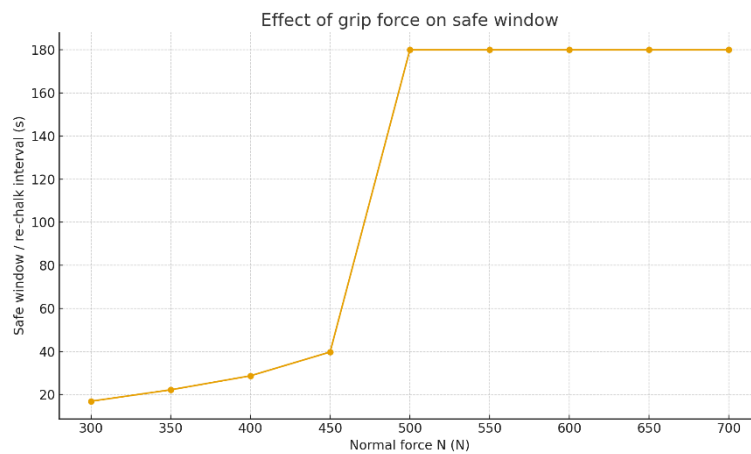


Figure 6.4.6 : Increasing N extends the window (since $S \propto \mu N$), with diminishing returns once moisture makes μ low. Useful to choose a realistic target grip squeeze.

6.5 Scenario set

- **S1—Baseline humid room:** $\varphi = 0.6$, still air, fresh chalk; report failure time and $h(t)$.
- **S2—Add a small fan:** same as S1 but $u = 0.5 \text{ m}\cdot\text{s}^{-1}$; compare k_m , failure time, and required re-chalk.
- **S3—No chalk:** $\theta_0 = 1$ and $M_{\text{cap}} \rightarrow 0$; quantify the loss of window.
- **S4—Partially used chalk:** $\theta_0 = 0.5$; show shortened interval.
- **S5—High sweater:** $J_{\text{sweat}} = 1 \times 10^{-3}$; test the value of fan vs chalk.

- **S6—Knurl/force sensitivity:** sweep N (normal force) and show how stronger squeeze extends the window by $\mu(h)N$.
- Scenario comparison (S1–S6) safety margin over time: We can see interventions side-by-side.

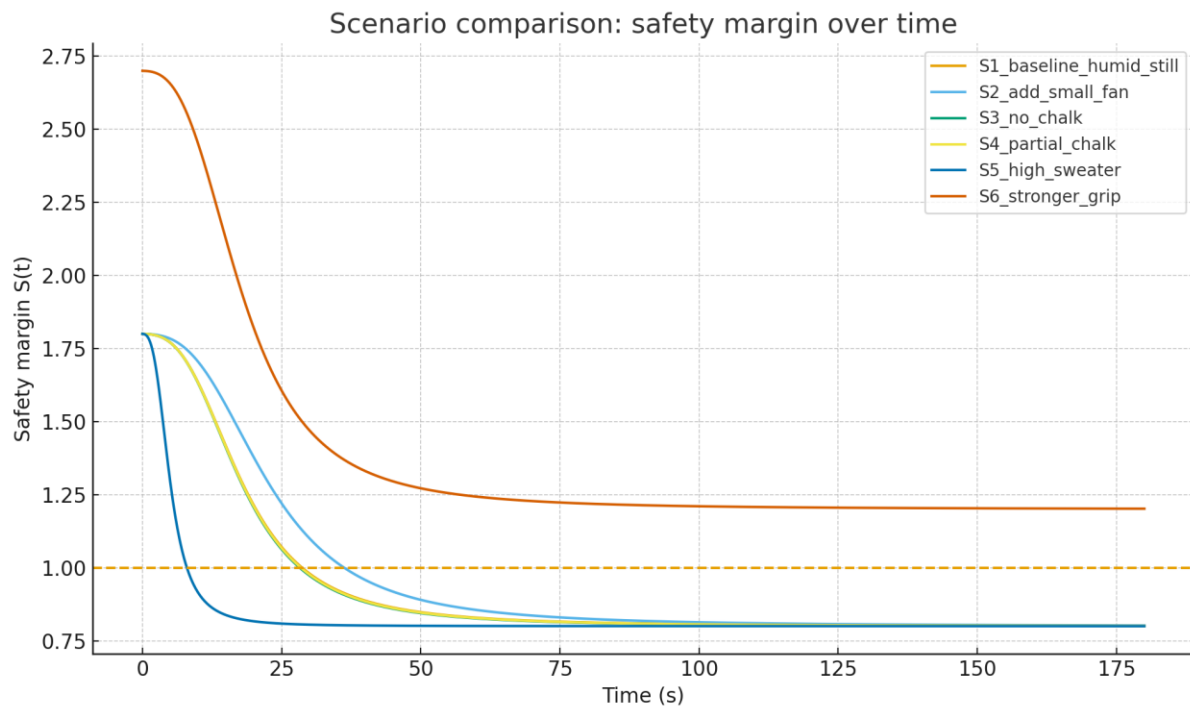


Figure 6.4.7: All six scenarios

Result:

	A	B	C	D	E	F	G	H	I	J	K	L
1	scenario	RH	u (m/s)	k_m (m/s)	J_{sweat} (kg)	M_{cap} (kg)	θ_0	N (N)	F_{req} (N)	failure_time	rechalk_time_s	
2	S1_baseline	0.6	0.05	0.002858	0.0003	0.005	0	400	200	28.7		
3	S2_add_small_fan	0.6	0.5	0.009038	0.0003	0.005	0	400	200	36.25		
4	S3_no_chalk	0.6	0.05	0.002858	0.0003	1.00E-12	1	400	200	28.25	0.05	
5	S4_partial_chalk	0.6	0.05	0.002858	0.0003	0.005	0.5	400	200	28.5		
6	S5_high_sweater	0.6	0.05	0.002858	0.001	0.005	0	400	200	8	130.2	
7	S6_stronger_grip	0.6	0.05	0.002858	0.0003	0.005	0	600	200			

Figure 6.4.8: Result and scenario table in excel

- S1 Baseline (RH=0.6, $u=0.05$ m/s, ($N=400$) N): 28.7 s to threshold. Chalk doesn't saturate in 180 s.
- S2 + Small fan ($u=0.5$ m/s): 36.25 s ($\approx +7.6$ s vs baseline). (k_m) rises from 2.86×10^{-3} to 9.04×10^{-3} m/s.
- S3 No chalk (capacity ≈ 0): 28.25 s. “Re-chalk” is effectively immediate (0.05 s, your time step), since θ starts at 1.

- S4 Partially used chalk ($\theta_0=0.5$): 28.5 s.
- S5 High sweater : 8.0 s. Re-chalk time listed (130.2 s) is irrelevant because failure happens first.
- S6 Stronger grip ((N=600) N): No failure within 180 s (margin stays > 1).

6.6 Validation and interpretation

We validate qualitative trends against the literature you shared:

- (i) more reps with chalk in CG-MTP (our model should predict a longer no-slip window with fresh chalk)
- (ii) context-dependent chalk benefit and improved consistency from fingertip tribology (our curves should show diminishing returns as chalk saturates and stronger effect in drier air). Any mismatches are noted and discussed.

All equations, parameters, and plots are produced from a single script (Python). Inputs are grouped at the top for easy editing (RH, air speed, chalk state, load). Figures are I pasted into this report

7. Conclusion

We turned a familiar gym problem—the *bar slipping after a few reps*—into a clean, quantitative model built from course ideas: a mass balance on the sweat film, a transfer-coefficient for evaporation, a finite-capacity chalk sink, and a moisture-to-friction link. By tracking film thickness $h(t)$, chalk saturation $\theta(t)$, friction $\mu(h)$, and the safety margin $S(t) = \mu(h)N/F_{\text{req}}$, the model gives two practical outputs: a no-slip window and a re-chalk interval.

The main lessons are straightforward. Humidity shortens grip time by reducing the evaporation driving force. Airflow helps a lot—even a small fan raises the air-side coefficient k_m and noticeably extends the safe window. Chalk helps most when fresh and then plateaus as it saturates, so re-apply on a schedule rather than over-using it. Squeezing harder (higher N) improves the margin but doesn't stop wetting, so ventilation and timely chalking remain the first controls. These trends match both training experience and published evidence on chalk's context-dependent friction effects.

There are limits: local hot spots, detailed knurl geometry, and exact friction calibration are simplified. Still, the guidance is clear—keep the interface dry to stay in the high- μ region. Practically: use light, regular chalking, add a small fan (especially on humid days), and wipe the bar; prefer a more aggressive knurl when grip is the limiter. The included Python script makes the analysis reproducible and easy to adapt to general gym conditions (RH, airflow, load), and with a short calibration session the same framework can become a simple, personalized “safe-seconds” chart for daily training.

8. Limitations and further work

We model the hand–bar contact as one uniform patch with a single film thickness $h(t)$. Local wet spots and knurl valleys can fail earlier. Environment within a set is held constant (T, RH, airflow), so mild cooling and short-term changes are ignored. Chalk is a simple porous sink with one uptake rate k_s and a finite capacity M_{cap} ; particle size, caking, and third-body shear are not resolved. The friction law $\mu(h)$ is a smooth, empirical dry→wet curve and not bar/skin calibrated. The lift demand F_{req} is treated as a representative value, not fully time-varying.

The uncertainties are Sweat rate, local airflow, chalk properties, skin oils/calluses, and bar coating all vary. We therefore use ranges and sensitivity checks rather than exact numbers.

We need to Calibrate the model by measuring friction on a bar segment under controlled RH/airflow and fit $\mu_{\text{dry}}, \mu_{\text{wet}}, h_c, n$; weigh hands or a chalk pad before/after timed holds to estimate J_{sweat} and M_{cap} ; and log local RH/airflow at normal gym to compare predicted re-chalk times with real reps. Improve physics with a spatial film $h(x, t)$, a light thermal balance (so k_m and $p_{ws}(T)$ evolve), a knurl parameter, and time-varying $F_{\text{req}}(t)$. Make it practical via a quick calculator (inputs: RH, airflow, load, last chalk → “safe seconds/reps” + re-chalk suggestion) and a one-page poster for the weight room.

9. Acknowledgement

I would like to thank Prof. G. K. Suraishkumar for the opportunity to carry out this CFA as part of BT5051 – Transport Phenomena in Biological Systems. The exercise helped me connect continuum-analysis ideas to a real problem from training and strengthened my modeling skills. I am also grateful to the IIT Madras weightlifting coach and fellow lifters for practical inputs on bar grip and chalk use, and to my family and friends for their constant support throughout the semester. All plots and simulations in this report were generated by me; photographs or figures from other sources are cited where used.

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Python code

```
import math
import os
from dataclasses import dataclass, asdict

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# ----- Helpers -----
def p_ws_Tetens(T_K: float) -> float:
    """Saturation vapor pressure (Pa), Magnus-Tetens form."""
    T_C = T_K - 273.15
    return 610.94 * math.exp((17.625 * T_C) / (T_C + 243.04))

def k_m_from_correlation(u, L, Dv, nu):
    """Air-side mass transfer coefficient (m/s) from Sherwood correlation."""
    Re = max(u * L / nu, 1e-12)
    Sc = max(nu / Dv, 1e-12)
    Sh = 0.664 * (Re ** 0.5) * (Sc ** (1.0 / 3.0))
    return Sh * Dv / L

# ----- Model Parameters -----
@dataclass
class Params:
    # Environment / geometry
    T: float = 298.0          # K
    RH: float = 0.60          # 0-1
    u: float = 0.05           # m/s (air speed near hand)
    L: float = 0.08           # m (contact length scale)
    Dv: float = 2.5e-5         # m²/s (H2O vapor in air)
    nu: float = 1.5e-5         # m²/s (air kinematic viscosity)
    # Liquid film
    rho_w: float = 1000.0     # kg/m³
    h0: float = 2e-7          # m (nearly dry start)
    # Sources/sinks
    J_sweat: float = 3e-4      # kg/m²/s
    k_s: float = 1.0          # 1/s (chalk uptake rate)
    M_cap: float = 5e-3       # kg/m² (chalk capacity)
    theta0: float = 0.0       # 0-1 (initial chalk saturation)
    # Friction law
    mu_dry: float = 0.9
    mu_wet: float = 0.4
    h_c: float = 5e-6         # m
    n_fr: float = 3.0
    # Mechanics

    N: float = 400.0          # N (normal force)
    F_req: float = 200.0      # N (tangential demand)
    S_min: float = 1.0        # safety threshold
    # Integration
    dt: float = 0.05          # s
    t_max: float = 180.0      # s
    # Chalk management
    theta_chalk_limit: float = 0.8 # re-chalk trigger

# ----- Fluxes and closures -----
def evaporation_flux(params: Params) -> float:
    """kg/m²/s, air-side controlled evaporation."""
    k_m = k_m_from_correlation(params.u, params.L, params.Dv, params.nu)
    Mw = 0.01801528          # kg/mol
    R = 8.314462618          # J/mol/K
    pws = p_ws_Tetens(params.T)
    c_vs = Mw * pws / (R * params.T)          # surface vapor conc.
    c_vinf = Mw * (params.RH * pws) / (R * params.T) # ambient vapor conc.
    return k_m * max(c_vs - c_vinf, 0.0)

def chalk_flux(h: float, theta: float, params: Params) -> float:
    """kg/m²/s, lumped chalk uptake with finite capacity."""
    return params.k_s * h * max(1.0 - theta, 0.0)

def mu_of_h(h: float, params: Params) -> float:
    """friction coefficient as a smooth function of film thickness."""
    return params.mu_wet + (params.mu_dry - params.mu_wet) / (1.0 + (h / params.h_c) ** params.n_fr)

# ----- Simulator -----
def simulate(params: Params):
    J_ev_base = evaporation_flux(params)
    k_m_val = k_m_from_correlation(params.u, params.L, params.Dv, params.nu)

    t_steps = int((params.t_max / params.dt) + 1)
    t = np.linspace(0.0, params.t_max, t_steps)
    h = np.zeros_like(t)
    theta = np.zeros_like(t)
    mu = np.zeros_like(t)
    S = np.zeros_like(t)

    h[0] = params.h0
    theta[0] = params.theta0
    mu[0] = mu_of_h(h[0], params)
    S[0] = (mu[0] * params.N) / max(params.F_req, 1e-12)

    failure_time = None
    rechalk_time = None

    def f_h(h_val, theta_val):
        return (params.J_sweat - J_ev_base - chalk_flux(h_val, theta_val, params)) / params.rho_w

    def f_theta(h_val, theta_val):
        return chalk_flux(h_val, theta_val, params) / max(params.M_cap, 1e-12)

    # 4th-order Runge-Kutta
    for i in range(1, t_steps):
        h_i = h[i - 1]
        th_i = theta[i - 1]
        dt = params.dt

        k1_h = f_h(h_i, th_i)
        k1_th = f_theta(h_i, th_i)

        k2_h = f_h(h_i + 0.5 * dt * k1_h, th_i + 0.5 * dt * k1_th)
        k2_th = f_theta(h_i + 0.5 * dt * k1_h, th_i + 0.5 * dt * k1_th)

        k3_h = f_h(h_i + 0.5 * dt * k2_h, th_i + 0.5 * dt * k2_th)
        k3_th = f_theta(h_i + 0.5 * dt * k2_h, th_i + 0.5 * dt * k2_th)

        k4_h = f_h(h_i + dt * k3_h, th_i + dt * k3_th)
        k4_th = f_theta(h_i + dt * k3_h, th_i + dt * k3_th)

        h[i] = max(h_i + (dt / 6.0) * (k1_h + 2 * k2_h + 2 * k3_h + k4_h), 0.0)
        theta[i] = min(max(th_i + (dt / 6.0) * (k1_th + 2 * k2_th + 2 * k3_th + k4_th), 0.0), 1.0)

    mu[i] = mu_of_h(h[i], params)
    S[i] = (mu[i] * params.N) / max(params.F_req, 1e-12)

    if failure_time is None and S[i] <= params.S_min:
        failure_time = t[i]
    if rechalk_time is None and theta[i] >= params.theta_chalk_limit:
        rechalk_time = t[i]

    if failure_time is None:
        failure_time = float('inf')
    if rechalk_time is None:
        rechalk_time = float('inf')

    return {
        "t": t, "h": h, "theta": theta, "mu": mu, "S": S,
        "failure_time": failure_time, "rechalk_time": rechalk_time,
        "k_m": k_m_val
    }

# ----- Plot helper -----
def save_and_show(fig, path):
    fig.tight_layout()
    fig.savefig(path, dpi=200, bbox_inches="tight")
    plt.show()

# ----- Main -----
if __name__ == "__main__":
    outdir = "."
    os.makedirs(outdir, exist_ok=True)

    # Scenarios
    base = Params()
    scenarios = {
        "S1_baseline_humid_still": base,
        "S2_add_small_fan": Params(**asdict(base), "u": 0.5)),
        "S3_no_chalk": Params(**asdict(base), "M_cap": 1e-12, "theta0": 1.0)),
        "S4_partial_chalk": Params(**asdict(base), "theta0": 0.5)),
        "S5_high_sweater": Params(**asdict(base), "J_sweat": 1e-3)),
        "S6_stronger_grip": Params(**asdict(base), "N": 600.0)),
    }

    # Run simulations
    results = {name: simulate(p) for name, p in scenarios.items()}

    # Summary CSV
    rows = []
    for name, pars in scenarios.items():
        res = results[name]
        rows.append({
            "scenario": name,
            "RH": pars.RH,
            "u (m/s)": pars.u,

            "k_m (m/s)": res["k_m"],
            "J_sweat (kg/m²/s)": pars.J_sweat,
            "M_cap (kg/m²)": pars.M_cap,
            "theta0": pars.theta0,
            "N (N)": pars.N,
            "F_req (N)": pars.F_req,
            "failure_time_s": None if math.isinf(res["failure_time"]) else res["failure_time"],
            "rechalk_time_s": None if math.isinf(res["rechalk_time"]) else res["rechalk_time"],
        })
    summary_df = pd.DataFrame(rows)
    summary_df.to_csv(os.path.join(outdir, "grip_summary.csv"), index=False)
    print("Saved grip_summary.csv")

    # Baseline time traces (S1)
    res1 = results["S1_baseline_humid_still"]
    t = res1["t"]

    fig = plt.figure()
    plt.plot(t, res1["h"])
    plt.xlabel("Time (s)")
    plt.ylabel("Film thickness h (m)")
    plt.title("Baseline (S1): Moisture film h(t)")
    save_and_show(fig, os.path.join(outdir, "S1_h_t.png"))
```

```

fig = plt.figure()
plt.plot(t, res1["theta"])
plt.xlabel("Time (s)")
plt.ylabel("Chalk saturation  $\theta$ ")
plt.title("Baseline (S1): Chalk saturation  $\theta(t)$ ")
save_and_show(fig, os.path.join(outdir, "S1_theta_t.png"))

fig = plt.figure()
plt.plot(t, res1["mu"])
plt.xlabel("Time (s)")
plt.ylabel("Friction coefficient  $\mu$ ")
plt.title("Baseline (S1): Friction  $\mu(t)$ ")
save_and_show(fig, os.path.join(outdir, "S1_mu_t.png"))

fig = plt.figure()
plt.plot(t, res1["S"])
plt.axhline(y=base.S_min, linestyle="--")
plt.xlabel("Time (s)")
plt.ylabel("Safety margin S(t)")
plt.title("Baseline (S1): Safety margin S(t)")
save_and_show(fig, os.path.join(outdir, "S1_S_t.png"))

# Re-chalk interval vs humidity (still vs fan)
RH_values = np.linspace(0.4, 0.8, 9)
t_fail_still = []
t_fail_fan = []
for RH in RH_values:
    pars_still = Params(**{"base": float(RH), "u": 0.05})
    pars_fan = Params(**{"base": float(RH), "u": 0.5})
    r_still = simulate(pars_still)
    r_fan = simulate(pars_fan)
    t_rc_still = min(r_still["failure_time"], r_still["rechalk_time"])
    t_rc_fan = min(r_fan["failure_time"], r_fan["rechalk_time"])
    t_fail_still.append(pars_still.t_max if math.isinf(t_rc_still) else t_rc_still)
    t_fail_fan.append(pars_fan.t_max if math.isinf(t_rc_fan) else t_rc_fan)

fig = plt.figure()
plt.plot(RH_values, t_fail_still, marker="o", label="Still air (~0.05 m/s)")
plt.plot(RH_values, t_fail_fan, marker="s", label="Small fan (~0.5 m/s)")
plt.xlabel("Relative Humidity (fraction)")
plt.ylabel("Suggested re-chalk interval (s)")
plt.title("Re-chalk interval vs humidity")
plt.legend()
save_and_show(fig, os.path.join(outdir, "rechalk_vs_humidity.png"))

# Safe window vs grip force N
N_values = np.linspace(300, 700, 9)
t_safe_N = []
for N in N_values:
    parsN = Params(**{"base": float(N)})
    rN = simulate(parsN)
    t_rc = min(rN["failure_time"], rN["rechalk_time"])
    t_safe_N.append(parsN.t_max if math.isinf(t_rc) else t_rc)

fig = plt.figure()
plt.plot(N_values, t_safe_N, marker="o")
plt.xlabel("Normal force N (N)")
plt.ylabel("Safe window / re-chalk interval (s)")
plt.title("Effect of grip force on safe window")
save_and_show(fig, os.path.join(outdir, "safe_window_vs_N.png"))

# Scenario comparison: safety margin time series
fig = plt.figure()
for name in ["S1_baseline_humid_still",
             "S2_add_small_fan",
             "S3_no_chalk",
             "S4_partial_chalk",
             "S5_high_sweater",
             "S6_stronger_grip"]:
    plt.plot(results[name]["t"], results[name]["S"], label=name)
plt.axhline(y=base.S_min, linestyle="--")
plt.xlabel("Time (s)")
plt.ylabel("Safety margin S(t)")
plt.title("Scenario comparison: safety margin over time")
plt.legend()
save_and_show(fig, os.path.join(outdir, "scenario_safety_margins.png"))

print("All figures saved in:", os.path.abspath(outdir))

```

Python code file :

https://colab.research.google.com/drive/1UZ5xjnyZZMNq_zjbfFKRthsek8LMHTii?usp=sharing

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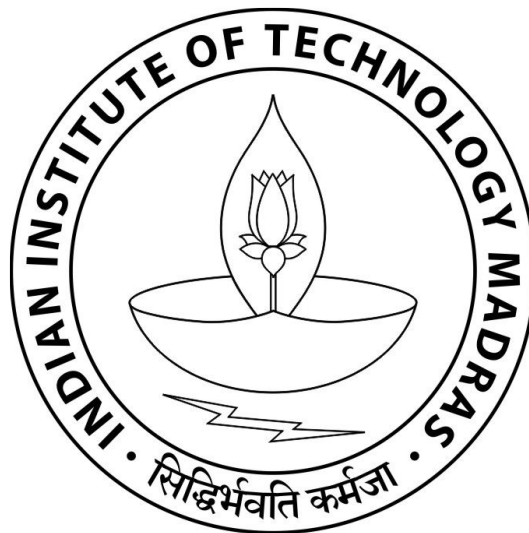
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Barbell Grip: Moisture, Chalk, and Slip Risk

Choose Focus Analysis Exercise

BT5051: Transport Phenomena in Biological Systems



Vimal Singh

BE23B042

3rd November, 2025

IIT MADRAS

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1. Introduction

Grip reliability is a practical safety and performance concern in strength training. As a set progresses, perspiration accumulates at the hand–bar interface, forming a thin moisture film that can substantially reduce friction and increase the likelihood of micro-slip or bar roll. Athletes commonly apply magnesium carbonate (chalk) to absorb moisture and improve feel, yet chalking practices are often ad hoc and sensitive to room humidity, airflow, bar knurling, and individual sweat rates. This project develops a concise, first-principles model to quantify how sweat, evaporation, and chalk jointly determine the time window over which a secure grip can be maintained.

The core idea is straightforward: the friction coefficient μ at the palm–bar interface is strongly dependent on the interfacial moisture state. Moisture increases through sweat secretion and decreases through two sinks—evaporation to room air and sorption into a finite-capacity chalk layer. When the moisture film is thin (dry regime), asperity interlock and adhesion support higher friction; as the film thickens (boundary-lubricated regime), μ drops and slip risk rises. Chalk improves the margin until it becomes saturated, after which its effectiveness declines.



Figure 1. Preparing hands with chalk (magnesium carbonate), a common strategy to reduce moisture and improve friction at the palm–bar interface.

Image source : <https://share.google/images/jtEkwZwcaedFq4jmo>



Figure 2. Illustration of the CG-MTP (clean-grip mid-thigh pull), a heavy barbell movement where grip reliability is critical.

Image source : <https://share.google/images/85PFB8SCzs0NevytC>

This report treats the hand–bar contact as a small, representative area and uses standard ideas from Continuum Analysis/Transport Phenomena—mass conservation, fluxes, and transfer-coefficient methods—to turn the above narrative into a compact, quantitative description of grip behavior during a set. With transparent, gym-realistic parameters (temperature, RH, local air speed, sweat rate, chalk amount, and normal force), we track how the moisture film evolves and how that state maps to friction and a simple safety margin. The focus is intentionally practical: to make clear, reproducible plots that show how long a secure grip lasts under typical gym room conditions, how ventilation or humidity shift that window, and why timely re-chalking helps. By grounding a familiar training issue in first principles, the analysis connects classroom concepts to a decision a lifter makes every day at the platform.

2. Motivation

Strength training has been a steady interest of mine, even though the last few semesters have been difficult. Between academic load and my father's blood cancer diagnosis in second semester, I stopped going to the gym regularly. The interest never went away, but time and motivation did. When I did manage NSO sessions at the IIT Madras weightlifting area, I kept running into the same practical problem: after just two or three reps, my palms would sweat, the knurled bar would feel slick, and the weight became harder to control. Applying chalk helped immediately—but I was guessing when to use it and how much.

This CFA gives me a constructive way to turn that lived experience into a clear, quantitative understanding. Instead of relying on trial and error, I want to explain why the grip “falls off” mid-set and how simple actions—re-chalking at the right time, improving airflow with a small fan, or choosing a different knurl—can prevent it. The topic also keeps me connected to the gym even while academics and family responsibilities limit training time; it channels my interest into analysis that can help me and others lift more safely.

The problem aligns naturally with this course. The moisture at the hand–bar interface can be treated with mass conservation (sweat in; evaporation and chalk absorption out) and transfer-coefficient ideas for air-side removal. A simple constitutive relation links that moisture state to friction, allowing us to predict when available friction drops below what the lift requires. These are exactly the tools emphasized in Continuum Analysis/Transport Phenomena, applied to a familiar, real-world setting.

Looking ahead, I want to become an AI/ML engineer, and my Nirmaan internships have trained me to turn intuition into code, structure messy problems, and validate models against reality. I will bring that mindset to this CFA—use transparent parameters, write a reproducible Python script, run sensitivity sweeps, and convert the outputs into simple, gym-facing recommendations. In that sense, this project is both a personal problem I care about and a small applied modeling exercise that builds the skills I aim to use in my career.

3. Objective

- Build a small, physics-based model of barbell grip that tracks moisture at the hand–bar interface from sweat, evaporation (airflow), and chalk.
- Link moisture to friction with a simple $\mu(h)$ relation so we can estimate grip strength over time.
- Predict the no-slip window (time/reps before failure) and the best re-chalk interval under typical gym conditions (humidity, airflow, knurl, load).

4. Concepts used

4.1 Conservation of mass

A mass balance on a small control area A at the palm–bar interface. A thin liquid film of thickness $h(t)$ covers this area; the areal mass of water is $\rho_w h (\text{kg} \cdot \text{m}^{-2})$. One source adds water (sweat from eccrine glands), and two sinks remove it (evaporation to the room air and absorption into chalk).

General form:

In + Generation – Out – Consumption = Accumulation. (4.1.1)

$$\mathbf{I} + \mathbf{G} - \mathbf{O} - \mathbf{C} = \mathbf{A}$$

Our film (areal form): one source (sweat) and two sinks (evaporation, chalk).

$$\frac{d(\rho_w h)}{dt} = J_{\text{sweat}} - J_{\text{evap}} - J_{\text{chalk}} \Rightarrow \boxed{\frac{dh}{dt} = \frac{1}{\rho_w} (J_{\text{sweat}} - J_{\text{evap}} - J_{\text{chalk}})} \quad (4.1.2)$$

- $h(t)$: film thickness (m), ρ_w : water density ($\text{kg} \cdot \text{m}^{-3}$), J : flux ($\text{kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$).
This ODE is the backbone of the model.

This ODE is the core of our model; later sections just define each flux. We assume a uniform thin film over area A (lumped model), which is reasonable for short sets focused on when slip begins, not where. We start nearly dry: $h(0) = h_0 \ll h_c$.

4.2 Mass flux & Fick's idea

Mass (or molar) flux is Amount crossing unit area per time, normal to the surface. For diffusion through a thin layer:

$$\boxed{J = -D \frac{\partial c}{\partial y}} \quad (4.2)$$

- D : diffusivity (m^2/s)
- c : concentration

- y : normal direction

where J is flux, D diffusivity (m^2/s), c concentration, and y the normal coordinate. With a finite gas-side boundary layer of thickness δ :

$$J \approx D \frac{c_{v,s} - c_{v,\infty}}{\delta} = k_m (c_{v,s} - c_{v,\infty}),$$

$$k_m \equiv D/\delta.$$

I assume the gas side controls and the liquid film is well mixed—a standard evaporation simplification—so we can estimate k_m from airflow/geometry (via Re , Sc , Sh).

4.3 Transfer-coefficient model for evaporation

Following the course's k -type approach, we express evaporation to room air as

$$J_{\text{evap}} = k_m (c_{v,s} - c_{v,\infty}), \quad (4.3)$$

with k_m the air-side mass-transfer coefficient (m/s). Concentrations are obtained from the ideal-gas relation $c_v = \frac{M_w}{RT} p_v$. For a surface at temperature T ,

$$c_{v,s} = \frac{M_w}{RT} p_{ws}(T), \quad c_{v,\infty} = \frac{M_w}{RT} \phi p_{ws}(T), \quad (4.4)$$

where $p_{ws}(T)$ is saturation vapor pressure and ϕ is relative humidity. Thus high ϕ shrinks the driving force, slowing drying. To estimate k_m we use a standard Sherwood correlation for external flow over a small patch (analogy to Nusselt for heat):

$$Sh = \frac{k_m L}{D_v} \approx 0.664 Re^{1/2} Sc^{1/3} \Rightarrow k_m \approx 0.664 \frac{D_v}{L} Re^{1/2} Sc^{1/3}, \quad (4.5)$$

with $Re = uL/\nu$ and $Sc = \nu/D_v$. Here u is local air speed near the hand, L characteristic hand width, ν air kinematic viscosity, and D_v the diffusivity of water vapor in air. This correlation makes the model actionable: a small fan increases

u , which increases Reand therefore k_m , enlarging the no-slip window. In code, k_m is computed once per scenario (constant T, φ, u during a set).

4.4 Chalk as a capacity-limited porous sink

Chalk (magnesium carbonate) is treated as a thin porous layer that absorbs liquid water until full.

- M_{cap} : max water per area the chalk can hold ($\text{kg}\cdot\text{m}^{-2}$).
- $m_c(t)$: water already stored per area; $\theta = m_c/M_{\text{cap}} \in [0,1]$ (saturation).

Uptake law (lumped):

$$\boxed{J_{\text{chalk}} = k_s h (1 - \theta)}, \quad \boxed{\frac{d\theta}{dt} = \frac{J_{\text{chalk}}}{M_{\text{cap}}}} \quad (4.6)$$

- $k_s(\text{s}^{-1})$ bundles porosity/permeability/contact effects.
As $\theta \rightarrow 1$, uptake $\rightarrow 0$: saturated chalk stops helping.

4.5 Friction–moisture relation

We model friction as a smooth drop from “dry” to “wet” with film thickness h :

$$\mu(h) = \mu_{\text{wet}} + \frac{\mu_{\text{dry}} - \mu_{\text{wet}}}{1 + (h/h_c)^n} \quad (4.7)$$

- $\mu_{\text{dry}}, \mu_{\text{wet}}$: friction limits (dry > wet).
- h_c : characteristic thickness where μ is halfway between dry and wet.
- n : steepness of the transition.
- N : normal force; available grip $F_{\text{avail}} = \mu(h)N$ (Coulomb).

thin films ($h \ll h_c$) $\rightarrow \mu \approx \mu_{\text{dry}}$;

thicker films ($h \gg h_c$) $\rightarrow \mu \approx \mu_{\text{wet}}$.

Squeezing harder raises μN , but only drying (chalk/airflow) moves μ back toward the dry, high-friction regime.

4.6 Failure (no-slip criterion and safety margin)

Available friction must exceed the lift's tangential demand:

$$F_{\text{avail}} = \mu(h(t)) N \quad (4.8)$$

Define a safety margin:

$$S(t) = \frac{\mu(h(t)) N}{F_{\text{req}}} \quad \text{fail when } S(t) \leq S_{\text{min}}. \quad (4.9)$$

We report the no-slip window as the time (or reps) until the threshold is crossed.

4.7 Non - dimensionalization (regime map for design)

Choose moisture scale h_c and an evaporation time scale

$$t_c = \frac{\rho_w h_c}{k_m \Delta c^*}, \Delta c^* = \frac{M_w}{RT} (p_{ws}(T_s) - \varphi p_{ws}(T_\infty)).$$

Let $h^* = h/h_c$, $t^* = t/t_c$. Then (4.1), (4.5)–(4.6) give

$$\frac{dh^*}{dt^*} = \frac{J_{\text{sweat}}}{k_m \Delta c^*} - 1 - \frac{k_s h_c}{k_m \Delta c^*} h^* (1 - \theta) \quad (4.10)$$

$$\frac{d\theta}{dt^*} = \frac{k_s \rho_w h_c}{M_{\text{cap}}} h^* (1 - \theta) \quad (4.11)$$

The groups mean (simple):

- Π_s — sweat input / evaporation capacity (Damköhler-like).
 - $\Pi_s \ll 1$: evaporation alone keeps the grip dry.

- $\Pi_s \gtrsim 1$: need chalk or stronger airflow.
- κ — chalk strength / evaporation strength.
 - Large κ : fresh chalk removes moisture quickly.
- β — how fast chalk saturates.
 - Large β : chalk fills fast $\rightarrow$ re-chalk more often.

These groups drive our regime maps and the re-chalk timing chart.

5. Overview of Relevant Information

5.1 Grip chalk in clean-grip mid-thigh pull (CG-MTP)

A study on high-school athletes compared CG-MTP performance at 70% 1-RM with and without magnesium carbonate. In a repeated-measures design, lifters completed more repetitions with chalk ($\approx 11.1 \pm 5.1$) than without ($\approx 9.1 \pm 3.1$), a statistically significant $\sim 22.5\%$ gain. The authors recommend chalk for lifts where grip can limit output. for our CFA, this supports treating chalk as a practical intervention that extends the “no-slip window,” which aligns with our plan to predict a re-chalk interval.

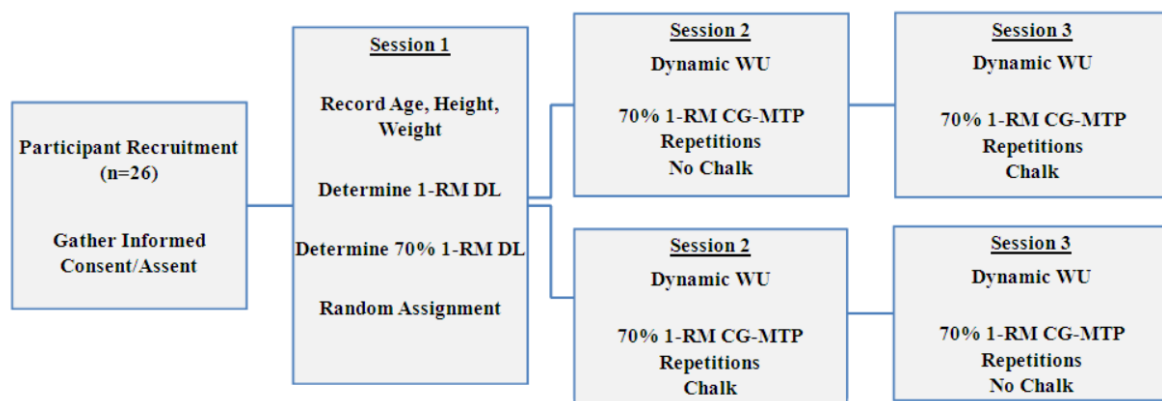


Figure 5.1. Study timeline of events. 1-RM one repetition maximum. CG-MTP closed grip midhigh pull

Image source :- <http://article.sapub.org/10.5923.j.sports.20201002.04.html>

5.2 Chalk as a friction modifier at the finger–surface interface

Fingertip tribology shows that chalk’s effect is context-dependent: it often helps in dry conditions by filling asperity gaps ($\uparrow$ real contact), but in wet conditions excess chalk can act as a sheared third body and lower friction; chalk also reduces variability, making grip more predictable. Skin friction arises from adhesion (real contact area) and deformation/ploughing; moisture first softens skin (can raise friction) and, when excessive, forms a lubricating film (lowers friction). We map these mechanisms into the model via film thickness $h(t)$, chalk saturation $\theta(t)$, and a moisture–friction law $\mu(h)$. The source is sweat flux J_{sweat} (grows h), while sinks are evaporation $J_{\text{evap}} = k_m(c_{v,s} - c_{v,\infty})$ (stronger with airflow; weaker at high RH) and chalk uptake $J_{\text{chalk}} = k_s h(1 - \theta)$ (effective until capacity M_{cap}). Design levers follow: improve ventilation ($\uparrow k_m$), manage

chalk freshness/amount ($\uparrow$ effective k_s , capacity), choose suitable knurl, and adjust normal force N . Accordingly, our figures report time-to-slip curves ($h(t)$, $\mu(h(t))$, margin $S(t)$), a re-chalk interval vs humidity/airflow/load, and sensitivities to knurl/roughness proxies and N —then we check that the trends align with CG-MTP results (more reps with chalk) and tribology findings (context-dependent benefit, improved consistency).

REFERENCE LINK : <https://eprints.whiterose.ac.uk/id/eprint/214893/10/clarke-et-al-2024-the-effectiveness-of-chalk-as-a-friction-modifier-for-finger-pad-contact-with-rocks-of-varying.pdf>

6. Approach

What we do. We build a small, reproducible simulation that tracks the water film at the palm–bar interface during a set, converts that moisture state into a friction coefficient, and reports the time (or reps) until slip risk crosses a chosen safety margin. All equations come from Section 4, and we keep parameters explicit so the plots can be regenerated for general gym conditions.

6.1 System definition and inputs

We model one hand's effective contact patch A on a knurled steel bar. The state variables are: film thickness $h(t)(m)$ and chalk saturation $\theta(t) \in [0,1]$. External inputs are:

- **Environment:** air temperature $T(K)$, relative humidity $\varphi(0-1)$, local air speed over the hand $u(m \cdot s^{-1})$.
- **Geometry/transport:** characteristic length L of the patch ($\sim$ hand width), vapor diffusivity D_v , air kinematic viscosity ν .
- **Chalk:** effective uptake rate $k_s(s^{-1})$, areal capacity $M_{cap}(kg \cdot m^{-2})$, initial saturation θ_0 .
- **Sweat:** flux $J_{sweat}(kg \cdot m^{-2} \cdot s^{-1})$; piecewise-constant (baseline $\rightarrow$ higher during the set).
- **Mechanics:** normal force $N(N)$ and tangential demand $F_{req}(N)$ representative of the lift phase we care about (steady hold or the “hardest” part of the pull).

We start each scenario with $h(0) = h_0$ (nearly dry after chalking/wiping) and $\theta(0) = \theta_0$ (fresh or partially used chalk).

6.2 Governing equations (implemented exactly as written)

The moisture balance on the control area is

$$\frac{dh}{dt} = \frac{1}{\rho_w} (J_{sweat} - J_{evap} - J_{chalk}), \rho_w \simeq 1000 \text{ kg} \cdot \text{m}^{-3}.$$

→ Constant film density ρ_w (liquid water, $\sim 1000 \text{ kg} \cdot \text{m}^{-3}$)

Evaporation uses the transfer-coefficient form

$$J_{\text{evap}} = k_m (c_{v,s} - c_{v,\infty})$$

$$k_m \approx 0.664 \frac{D_v}{L} \text{Re}^{\frac{1}{2}} \text{Sc}^{\frac{1}{3}}$$

$$\text{Re} = \frac{uL}{\nu}, \text{Sc} = \frac{\nu}{D_v}.$$

With ideal-gas relations:

$$c_{v,s} = \frac{M_w}{RT} p_{ws}(T)$$

$$c_{v,\infty} = \frac{M_w}{RT} \phi p_{ws}(T).$$

Chalk uptake is a capacity-limited sink:

$$J_{\text{chalk}} = k_s h (1 - \theta), \frac{d\theta}{dt} = \frac{J_{\text{chalk}}}{M_{\text{cap}}}.$$

Grip friction follows a smooth dry→wet relation:

$$\mu(h) = \mu_{\text{wet}} + \frac{\mu_{\text{dry}} - \mu_{\text{wet}}}{1 + (h/h_c)^n}.$$

Safety margin and failure time:

$$S(t) = \frac{\mu(h(t)) N}{F_{\text{req}}}, \text{declare "failure" when } S(t) \leq S_{\text{min}} \text{ (e.g., 1.0 or 1.2).}$$

6.3 Parameterization for base runs

We use transparent, gym-realistic starting values and then perform sensitivity sweeps:

- **Environment:** $T = 298\text{K}$ (25°C); $\varphi = 0.6$ (humid room) $\rightarrow$ also test 0.4 and 0.8. Air speed u : still air $0.05\text{ m}\cdot\text{s}^{-1}$; small fan $0.5\text{ m}\cdot\text{s}^{-1}$. $L = 0.08\text{m}$; $D_v = 2.5 \times 10^{-5}\text{m}^2/\text{s}$; $\nu = 1.5 \times 10^{-5}\text{m}^2/\text{s}$. These give $k_m \approx 1-9 \times 10^{-3}\text{m/s}$ (still $\rightarrow$ fan).
- **Sweat:** $J_{\text{sweat}} = 3 \times 10^{-4}\text{kg}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ during the set ($\pm 3\times$ in sweeps).
- **Chalk:** $k_s = 1.0\text{s}^{-1}$ (baseline), $M_{\text{cap}} = 5 \times 10^{-3}\text{kg}\cdot\text{m}^{-2}$, $\theta_0 = 0$ (fresh) or 0.5 (partially used).
- **Friction law:** $\mu_{\text{dry}} = 0.9$, $\mu_{\text{wet}} = 0.4$, $h_c = 5 \times 10^{-6}\text{m}$, $n = 3$.
- **Mechanics:** $N = 300-600\text{ N}$ per hand (firm squeeze); $F_{\text{req}} = 150-300\text{ N}$ per hand (depends on lift and moment arm). We report results as a margin $S(t)$ so users can map to their loads.

All values are editable in code and will be listed in a small table. And the python code I provided in the last of the CFA

6.4 Numerical solution and outputs

We integrate the coupled ODEs for $h(t)$ and $\theta(t)$ with a fixed-step 4th-order Runge–Kutta that I learnt in BT2020 course (time step $\Delta t = 0.05-0.1\text{ s}$; horizon 60–180 s). At each step we compute $\mu(h(t))$ and $S(t)$. Primary outputs:

1. **Time traces:** $h(t)$, $\theta(t)$, $\mu(h(t))$, and $S(t)$ (with the failure time marked).
2. **Re-chalk interval:** the earlier of (i) $S(t)$ crossing S_{min} or (ii) chalk saturation reaching $\theta \geq 0.8$.
3. **Design charts:** (a) re-chalk interval vs humidity and fan speed; (b) safe window vs N and F_{req} ; (c) sensitivity of outcomes to k_s and M_{cap} .
4. **Regime map:** contours of the non-dimensional groups Π_s , κ , β showing where evaporation alone suffices, where fresh chalk is effective, and where chalk saturates quickly.

- Baseline film thickness $h(t)$: You can see how quickly moisture builds on the palm–bar patch under baseline gym conditions.

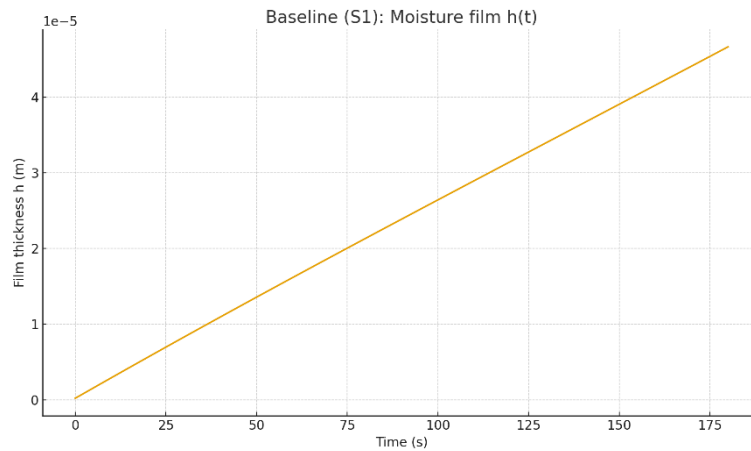


Figure 6.4.1: $h(t)$ starts near-dry and rises as sweat outpaces evaporation, then slows as chalk saturates. This is the upstream driver of all later effects (friction drop, slip risk).

- Baseline chalk saturation $\theta(t)$: Track how fast chalk fills up.

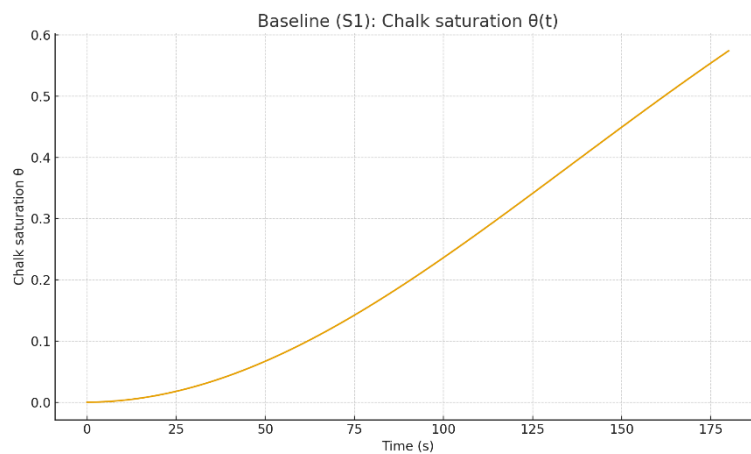


Figure 6.4.2 : Rapid early uptake ($\theta \uparrow$) then a plateau near $\theta \rightarrow 1$. The re-chalk time is chosen before saturation (e.g., $\theta \approx 0.8$), because benefit fades after that.

- Baseline friction $\mu(t)$: Convert moisture into grip quality.

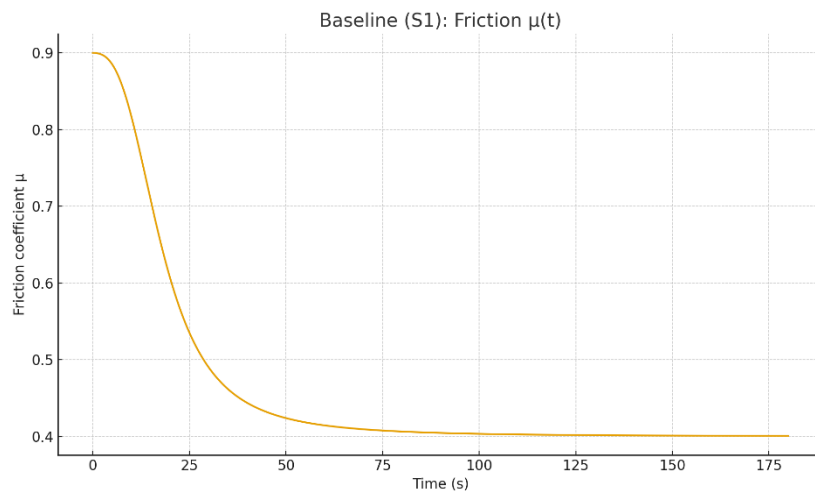


Figure 6.4.3: A smooth fall from μ_{dry} to μ_{wet} as $h(t)$ grows. The time where $\mu(t)$ bends downward corresponds to $h \sim h_c$ (onset of “wet” behavior).

- Baseline safety margin $S(t)$: This is the decision signal.

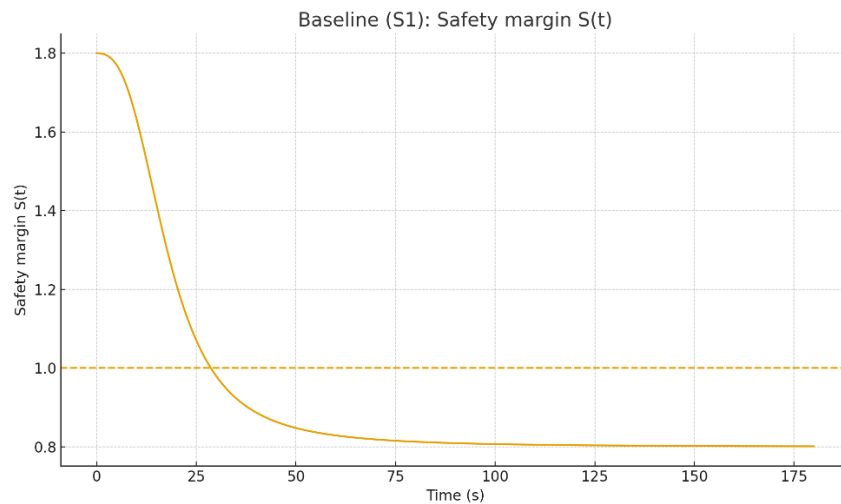


Figure 6.4.4 : $S(t) = \mu(h(t))N/F_{\text{req}}$ declining over time; the **no-slip window** is where $S(t)$ first crosses the threshold (S_{min}). That crossing gives the recommended **re-chalk interval** if earlier than chalk saturation.

- Re-chalk interval vs humidity (still air vs small fan): A design chart for the room environment.

Result : Partially used chalk ($\theta_0 = 0.5$): ≈ 28.5 s.

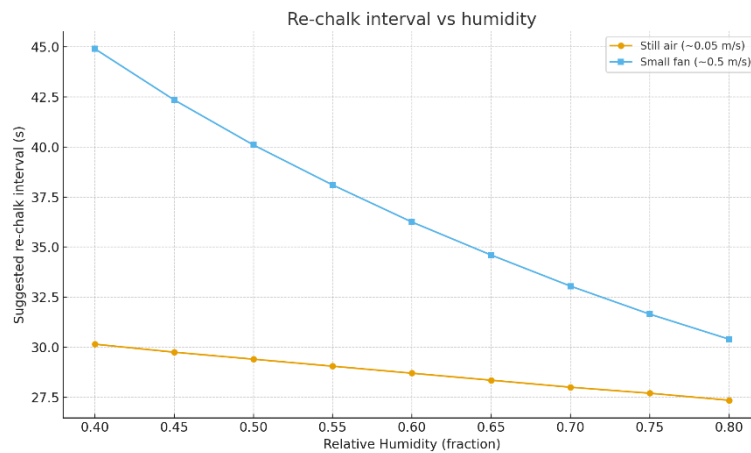


Figure 6.4.5 : Higher RH $\rightarrow$ shorter safe window. A small fan (higher k_m) shifts the curve upward (more seconds before re-chalk). Quantifies the benefit of ventilation on humid days.

- Safe window vs grip force N : How much does squeezing harder help.

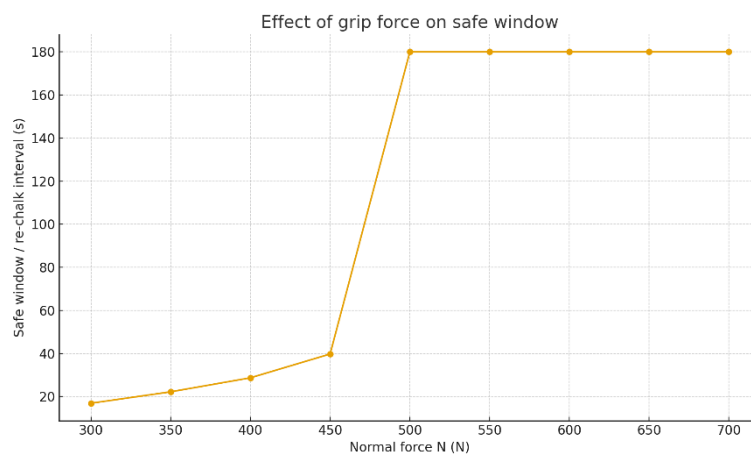


Figure 6.4.6 : Increasing N extends the window (since $S \propto \mu N$), with diminishing returns once moisture makes μ low. Useful to choose a realistic target grip squeeze.

6.5 Scenario set

- **S1—Baseline humid room:** $\phi = 0.6$, still air, fresh chalk; report failure time and $h(t)$.
- **S2—Add a small fan:** same as S1 but $u = 0.5 \text{ m} \cdot \text{s}^{-1}$; compare k_m , failure time, and required re-chalk.
- **S3—No chalk:** $\theta_0 = 1$ and $M_{\text{cap}} \rightarrow 0$; quantify the loss of window.
- **S4—Partially used chalk:** $\theta_0 = 0.5$; show shortened interval.
- **S5—High sweater:** $J_{\text{sweat}} = 1 \times 10^{-3}$; test the value of fan vs chalk.

- **S6—Knurl/force sensitivity:** sweep N (normal force) and show how stronger squeeze extends the window by $\mu(h)N$.
- Scenario comparison (S1–S6) safety margin over time: We can see interventions side-by-side.

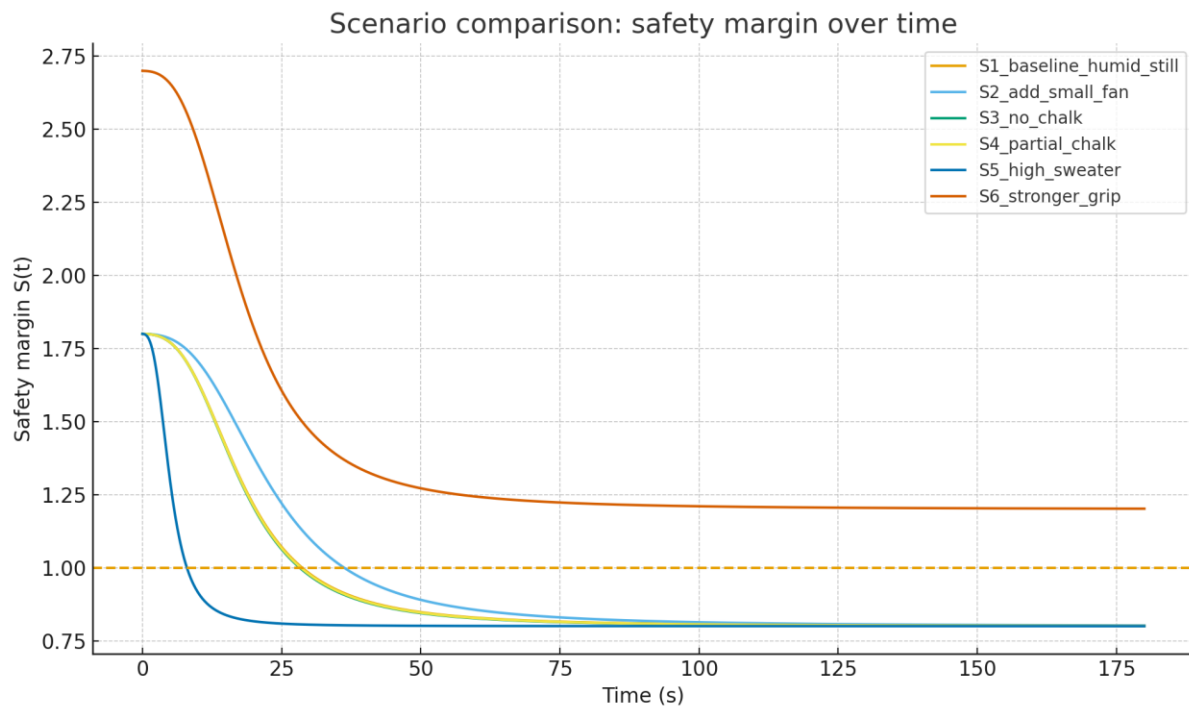


Figure 6.4.7: All six scenarios

Result:

	A	B	C	D	E	F	G	H	I	J	K	L
1	scenario	RH	u (m/s)	k _m (m/s)	J _{sweat} (kg)	M _{cap} (kg)	theta0	N (N)	F _{req} (N)	failure_time	rechalk_time_s	
2	S1_baseline	0.6	0.05	0.002858	0.0003	0.005	0	400	200	28.7		
3	S2_add_small_fan	0.6	0.5	0.009038	0.0003	0.005	0	400	200	36.25		
4	S3_no_chalk	0.6	0.05	0.002858	0.0003	1.00E-12	1	400	200	28.25	0.05	
5	S4_partial_chalk	0.6	0.05	0.002858	0.0003	0.005	0.5	400	200	28.5		
6	S5_high_sweater	0.6	0.05	0.002858	0.001	0.005	0	400	200	8	130.2	
7	S6_stronger_grip	0.6	0.05	0.002858	0.0003	0.005	0	600	200			

Figure 6.4.8: Result and scenario table in excel

- S1 Baseline (RH=0.6, $u=0.05$ m/s, ($N=400$) N): 28.7 s to threshold. Chalk doesn't saturate in 180 s.
- S2 + Small fan ($u=0.5$ m/s): 36.25 s ($\approx +7.6$ s vs baseline). (k_m) rises from 2.86×10^{-3} to 9.04×10^{-3} m/s.
- S3 No chalk (capacity ≈ 0): 28.25 s. “Re-chalk” is effectively immediate (0.05 s, your time step), since θ starts at 1.

- S4 Partially used chalk ($\theta_0=0.5$): 28.5 s.
- S5 High sweater : 8.0 s. Re-chalk time listed (130.2 s) is irrelevant because failure happens first.
- S6 Stronger grip ((N=600) N): No failure within 180 s (margin stays > 1).

6.6 Validation and interpretation

We validate qualitative trends against the literature you shared:

- (i) more reps with chalk in CG-MTP (our model should predict a longer no-slip window with fresh chalk)
- (ii) context-dependent chalk benefit and improved consistency from fingertip tribology (our curves should show diminishing returns as chalk saturates and stronger effect in drier air). Any mismatches are noted and discussed.

All equations, parameters, and plots are produced from a single script (Python). Inputs are grouped at the top for easy editing (RH, air speed, chalk state, load). Figures are I pasted into this report

7. Conclusion

We turned a familiar gym problem—the *bar slipping after a few reps*—into a clean, quantitative model built from course ideas: a mass balance on the sweat film, a transfer-coefficient for evaporation, a finite-capacity chalk sink, and a moisture-to-friction link. By tracking film thickness $h(t)$, chalk saturation $\theta(t)$, friction $\mu(h)$, and the safety margin $S(t) = \mu(h)N/F_{\text{req}}$, the model gives two practical outputs: a no-slip window and a re-chalk interval.

The main lessons are straightforward. Humidity shortens grip time by reducing the evaporation driving force. Airflow helps a lot—even a small fan raises the air-side coefficient k_m and noticeably extends the safe window. Chalk helps most when fresh and then plateaus as it saturates, so re-apply on a schedule rather than over-using it. Squeezing harder (higher N) improves the margin but doesn't stop wetting, so ventilation and timely chalking remain the first controls. These trends match both training experience and published evidence on chalk's context-dependent friction effects.

There are limits: local hot spots, detailed knurl geometry, and exact friction calibration are simplified. Still, the guidance is clear—keep the interface dry to stay in the high- μ region. Practically: use light, regular chalking, add a small fan (especially on humid days), and wipe the bar; prefer a more aggressive knurl when grip is the limiter. The included Python script makes the analysis reproducible and easy to adapt to general gym conditions (RH, airflow, load), and with a short calibration session the same framework can become a simple, personalized “safe-seconds” chart for daily training.

8. Limitations and further work

We model the hand–bar contact as one uniform patch with a single film thickness $h(t)$. Local wet spots and knurl valleys can fail earlier. Environment within a set is held constant (T, RH, airflow), so mild cooling and short-term changes are ignored. Chalk is a simple porous sink with one uptake rate k_s and a finite capacity M_{cap} ; particle size, caking, and third-body shear are not resolved. The friction law $\mu(h)$ is a smooth, empirical dry→wet curve and not bar/skin calibrated. The lift demand F_{req} is treated as a representative value, not fully time-varying.

The uncertainties are Sweat rate, local airflow, chalk properties, skin oils/calluses, and bar coating all vary. We therefore use ranges and sensitivity checks rather than exact numbers.

We need to Calibrate the model by measuring friction on a bar segment under controlled RH/airflow and fit $\mu_{\text{dry}}, \mu_{\text{wet}}, h_c, n$; weigh hands or a chalk pad before/after timed holds to estimate J_{sweat} and M_{cap} ; and log local RH/airflow at normal gym to compare predicted re-chalk times with real reps. Improve physics with a spatial film $h(x, t)$, a light thermal balance (so k_m and $p_{ws}(T)$ evolve), a knurl parameter, and time-varying $F_{\text{req}}(t)$. Make it practical via a quick calculator (inputs: RH, airflow, load, last chalk → “safe seconds/reps” + re-chalk suggestion) and a one-page poster for the weight room.

9. Acknowledgement

I would like to thank Prof. G. K. Suraishkumar for the opportunity to carry out this CFA as part of BT5051 – Transport Phenomena in Biological Systems. The exercise helped me connect continuum-analysis ideas to a real problem from training and strengthened my modeling skills. I am also grateful to the IIT Madras weightlifting coach and fellow lifters for practical inputs on bar grip and chalk use, and to my family and friends for their constant support throughout the semester. All plots and simulations in this report were generated by me; photographs or figures from other sources are cited where used.

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Python code

```

import math
import os
from dataclasses import dataclass, asdict

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# ----- Helpers -----
def p_ws_Tetens(T_K: float) -> float:
    """Saturation vapor pressure (Pa), Magnus-Tetens form."""
    T_C = T_K - 273.15
    return 610.94 * math.exp((17.625 * T_C) / (T_C + 243.04))

def k_m_from_correlation(u, L, Dv, nu):
    """Air-side mass transfer coefficient (m/s) from Sherwood correlation."""
    Re = max(u * L / nu, 1e-12)
    Sc = max(Dv / Dv, 1e-12)
    Sh = 0.664 * (Re ** 0.5) * (Sc ** (1.0 / 3.0))
    return Sh * Dv / L

# ----- Model Parameters -----
@dataclass
class Params:
    # Environment / geometry
    T: float = 298.0          # K
    RH: float = 0.60          # 0-1
    u: float = 0.05           # m/s (air speed near hand)
    L: float = 0.08           # m (contact length scale)
    Dv: float = 2.5e-5         # m^2/s (H2O vapor in air)
    nu: float = 1.5e-5         # m^2/s (air kinematic viscosity)
    # Liquid film
    rho_w: float = 1000.0     # kg/m^3
    h0: float = 2e-7          # m (nearly dry start)
    # Sources/sinks
    J_sweat: float = 3e-4     # kg/m^2/s
    k_s: float = 1.0          # 1/s (chalk uptake rate)
    M_cap: float = 5e-3       # kg/m^2 (chalk capacity)
    theta0: float = 0.0       # 0-1 (initial chalk saturation)
    # Friction law
    mu_dry: float = 0.9
    mu_wet: float = 0.4
    h_c: float = 5e-6         # m
    n_fr: float = 3.0         # Mechanics

    N: float = 400.0          # N (normal force)
    F_req: float = 200.0      # N (tangential demand)
    S_min: float = 1.0        # safety threshold

    # Integration
    dt: float = 0.05          # s
    t_max: float = 180.0      # s
    # Chalk management
    theta_chalk_limit: float = 0.8 # re-chalk trigger

# ----- Fluxes and closures -----
def evaporation_flux(params: Params) -> float:
    """kg/m^2/s, air-side controlled evaporation."""
    k_m = k_m_from_correlation(params.u, params.L, params.Dv, params.nu)
    Mw = 0.01801528          # kg/mol
    R = 8.314462618          # J/mol/K
    pws = p_ws_Tetens(params.T)
    c_vs = Mw * pws / (R * params.T)          # surface vapor conc.
    c_vinf = Mw * (params.RH * pws) / (R * params.T) # ambient vapor conc.
    return k_m * max(c_vs - c_vinf, 0.0)

def chalk_flux(h: float, theta: float, params: Params) -> float:
    """kg/m^2/s, lumped chalk uptake with finite capacity."""
    return params.k_s * h * max(1.0 - theta, 0.0)

def mu_of_h(h: float, params: Params) -> float:
    """Friction coefficient as a smooth function of film thickness."""
    return params.mu_wet * (params.mu_dry - params.mu_wet) / (1.0 + (h / params.h_c) ** params.n_fr)

# ----- Simulator -----
def simulate(params: Params):
    J_ev_base = evaporation_flux(params)
    k_m_val = k_m_from_correlation(params.u, params.L, params.Dv, params.nu)

    t_steps = int((params.t_max / params.dt) + 1)
    t = np.linspace(0.0, params.t_max, t_steps)
    h = np.zeros_like(t)
    theta = np.zeros_like(t)
    mu = np.zeros_like(t)
    S = np.zeros_like(t)

    h[0] = params.h0
    theta[0] = params.theta0
    mu[0] = mu_of_h(h[0], params)
    S[0] = (mu[0] * params.N) / max(params.F_req, 1e-12)

    failure_time = None
    rechalk_time = None

    def f_h(h_val, theta_val):
        return (params.J_sweat - J_ev_base - chalk_flux(h_val, theta_val, params)) / params.rho_w

    def f_theta(h_val, theta_val):
        return chalk_flux(h_val, theta_val, params) / max(params.M_cap, 1e-12)

    # 4th-order Runge-Kutta
    for i in range(1, t_steps):
        h_i = h[i-1]
        th_i = theta[i-1]
        dt = params.dt

        k1_h = f_h(h_i, th_i)
        k1_th = f_theta(h_i, th_i)

        k2_h = f_h(h_i + 0.5 * dt * k1_h, th_i + 0.5 * dt * k1_th)
        k2_th = f_theta(h_i + 0.5 * dt * k1_h, th_i + 0.5 * dt * k1_th)

        k3_h = f_h(h_i + 0.5 * dt * k2_h, th_i + 0.5 * dt * k2_th)
        k3_th = f_theta(h_i + 0.5 * dt * k2_h, th_i + 0.5 * dt * k2_th)

        k4_h = f_h(h_i + dt * k3_h, th_i + dt * k3_th)
        k4_th = f_theta(h_i + dt * k3_h, th_i + dt * k3_th)

        h[i] = h_i + (dt / 6.0) * (k1_h + 2 * k2_h + 2 * k3_h + k4_h)
        theta[i] = min(max(th_i + (dt / 6.0) * (k1_th + 2 * k2_th + 2 * k3_th + k4_th), 0.0), 1.0)

    mu[i] = mu_of_h(h[i], params)
    S[i] = (mu[i] * params.N) / max(params.F_req, 1e-12)

    if failure_time is None and S[i] <= params.S_min:
        failure_time = t[i]
    if rechalk_time is None and theta[i] >= params.theta_chalk_limit:
        rechalk_time = t[i]

    if failure_time is None:
        failure_time = float('inf')
    if rechalk_time is None:
        rechalk_time = float('inf')

    return {
        "t": t, "h": h, "theta": theta, "mu": mu, "S": S,
        "failure_time": failure_time, "rechalk_time": rechalk_time,
        "k_m": k_m_val
    }

# ----- Plot helper -----
def save_and_show(fig, path):
    fig.tight_layout()
    fig.savefig(path, dpi=200, bbox_inches="tight")
    plt.show()

# ----- Main -----
if __name__ == "__main__":
    outdir = "."
    os.makedirs(outdir, exist_ok=True)

    # Scenarios
    base = Params()
    scenarios = {
        "S1_baseline_humid_still": base,
        "S2_add_small_fan": Params(**asdict(base), "u": 0.5),
        "S3_no_chalk": Params(**asdict(base), "M_cap": 1e-12, "theta0": 1.0),
        "S4_partial_chalk": Params(**asdict(base), "theta0": 0.5),
        "S5_high_sweater": Params(**asdict(base), "J_sweat": 1e-3),
        "S6_stronger_grip": Params(**asdict(base), "N": 600.0),
    }

    # Run simulations
    results = {name: simulate(p) for name, p in scenarios.items()}

    # Summary CSV
    rows = []
    for name, pars in scenarios.items():
        res = results[name]
        rows.append({
            "scenario": name,
            "RH": pars.RH,
            "u (m/s)": pars.u,

            "k_m (m/s)": res["k_m"],
            "J_sweat (kg/m^2/s)": pars.J_sweat,
            "M_cap (kg/m^2)": pars.M_cap,
            "theta0": pars.theta0,
            "N (N)": pars.N,
            "F_req (N)": pars.F_req,
            "failure_time_s": None if math.isinf(res["failure_time"]) else res["failure_time"],
            "rechalk_time_s": None if math.isinf(res["rechalk_time"]) else res["rechalk_time"],
        })

    summary_df = pd.DataFrame(rows)
    summary_df.to_csv(os.path.join(outdir, "grip_summary.csv"), index=False)
    print("Saved grip_summary.csv")

    # Baseline time traces (S1)
    res1 = results["S1_baseline_humid_still"]
    t = res1["t"]

    fig = plt.figure()
    plt.plot(t, res1["h"])
    plt.xlabel("Time (s)")
    plt.ylabel("Film thickness h (m)")
    plt.title("Baseline (S1): Moisture film h(t)")
    save_and_show(fig, os.path.join(outdir, "S1_h_t.png"))

```

```
fig = plt.figure()
plt.plot(t, res1["theta"])
plt.xlabel("Time (s)")
plt.ylabel("Chalk saturation  $\theta$ ")
plt.title("Baseline (S1): Chalk saturation  $\theta(t)$ ")
save_and_show(fig, os.path.join(outdir, "S1_theta_t.png"))

fig = plt.figure()
plt.plot(t, res1["mu"])
plt.xlabel("Time (s)")
plt.ylabel("Friction coefficient  $\mu$ ")
plt.title("Baseline (S1): Friction  $\mu(t)$ ")
save_and_show(fig, os.path.join(outdir, "S1_mu_t.png"))

fig = plt.figure()
plt.plot(t, res1["S"])
plt.axhline(y=base.S_min, linestyle="--")
plt.xlabel("Time (s)")
plt.ylabel("Safety margin S(t)")
plt.title("Baseline (S1): Safety margin S(t)")
save_and_show(fig, os.path.join(outdir, "S1_S_t.png"))

# Re-chalk interval vs humidity (still vs fan)
RH_values = np.linspace(0.4, 0.8, 9)
t_fail_still = []
t_fail_fan = []
for RH in RH_values:
    pars_still = Params(**{"asdict(base)", "RH": float(RH), "u": 0.05})
    pars_fan = Params(**{"asdict(base)", "RH": float(RH), "u": 0.5})
    r_still = simulate(pars_still)
    r_fan = simulate(pars_fan)
    t_rc_still = min(r_still["failure_time"], r_still["rechalk_time"])
    t_rc_fan = min(r_fan["failure_time"], r_fan["rechalk_time"])
    t_fail_still.append(pars_still.t_max if math.isinf(t_rc_still) else t_rc_still)
    t_fail_fan.append(pars_fan.t_max if math.isinf(t_rc_fan) else t_rc_fan)

fig = plt.figure()
plt.plot(RH_values, t_fail_still, marker="o", label="Still air (~0.05 m/s)")
plt.plot(RH_values, t_fail_fan, marker="s", label="Small fan (~0.5 m/s)")
plt.xlabel("Relative Humidity (fraction)")
plt.ylabel("Suggested re-chalk interval (s)")
plt.title("Re-chalk interval vs humidity")
plt.legend()
save_and_show(fig, os.path.join(outdir, "rechalk_vs_humidity.png"))

# Safe window vs grip force N
N_values = np.linspace(300, 700, 9)
t_safe_N = []
for N in N_values:
    parsN = Params(**{"asdict(base)", "N": float(N)})
    rN = simulate(parsN)
    t_rc = min(rN["failure_time"], rN["rechalk_time"])
    t_safe_N.append(parsN.t_max if math.isinf(t_rc) else t_rc)

fig = plt.figure()
plt.plot(N_values, t_safe_N, marker="o")
plt.xlabel("Normal force N (N)")
plt.ylabel("Safe window / re-chalk interval (s)")
plt.title("Effect of grip force on safe window")
save_and_show(fig, os.path.join(outdir, "safe_window_vs_N.png"))

# Scenario comparison: safety margin time series
fig = plt.figure()
for name in ["S1_baseline_humid_still",
             "S2_add_small_fan",
             "S3_no_chalk",
             "S4_partial_chalk",
             "S5_high_sweater",
             "S6_stronger_grip"]:
    plt.plot(results[name]["t"], results[name]["S"], label=name)
plt.axhline(y=base.S_min, linestyle="--")
plt.xlabel("Time (s)")
plt.ylabel("Safety margin S(t)")
plt.title("Scenario comparison: safety margin over time")
plt.legend()
save_and_show(fig, os.path.join(outdir, "scenario_safety_margins.png"))

print("All figures saved in:", os.path.abspath(outdir))
```

Python code file :

https://colab.research.google.com/drive/1UZ5xjnyZZMNq_zjbfFKRthsek8LMHTii?usp=sharing